

# Shibli Istiak Alam

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## Research Interests

- Unsteady Aerodynamics
- Particle Image Velocimetry (PIV)
- Vortex Dynamics
- Low-Reynolds-Number Flows
- Bio-Inspired UAV Aerodynamics
- Gust Response

## Research Experience

**MS Thesis Research — Wake Dynamics of a NACA0012 Airfoil Undergoing a Single Oscillation at Low Reynolds Number.**  
*Clarkson University | Aug 2023 – Present | Advisor: Dr. Douglas G. Bohl*

- Conducted 2D PIV experiments on a NACA0012 airfoil undergoing single-cycle sinusoidal pitching at  $Re_c \approx 5,200$ , systematically varying Strouhal number and reduced frequency to isolate wake-formation mechanisms.
- Quantified vortex strength and structure (peak vorticity, circulation, core radius, convection speed) across 20+ kinematic cases, identifying pitch amplitude — not frequency — as the dominant driver of vortex strength.
- Developed a comparative framework linking isolated single-cycle wakes to continuous periodic pitching, demonstrating how wake-history effects progressively alter vortex evolution under repeated forcing.
- Operated and maintained a recirculating water tunnel and 2D PIV system, including laser alignment, camera calibration, seeding optimization, and troubleshooting synchronization/timing issues to ensure high-quality velocity field data.

## Education

**Clarkson University, Master of Science in Mechanical Engineering**

Aug 2023 – Present

- GPA: 3.83/4.0
- Relevant Coursework: Advanced Fluid Mechanics, Special Topics in Fluid Mechanics, Advanced Turbulence, Computational Fluid Dynamics, Deep Learning.

**Khulna University of Engineering & Technology, Bachelor of Science in Mechanical Engineering**

Feb 2017 – May 2022

- Thesis: Numerical Study on Heat Transfer in an Enclosure Filled with a Nanofluid and Containing a Heat Source at the Center.

## Publications

### Manuscripts in Preparation

- Shibli Istiak Alam, D. G. Bohl, "Wake Dynamics of a NACA0012 Airfoil Undergoing a Single Oscillation at Low Reynolds Number" (*journal manuscript in preparation*)

### Conference Publications

- SI Alam, SB Suja, MR Islam, "Analysis of Slot Jet impingement on a Moving Surface: A Numerical Approach", 7th International Conference on Mechanical, Industrial & Energy Engineering (ICMIEE-2022), Khulna, Bangladesh
- SI Alam, SB Suja, MS Islam, "Influence of Venturi on Heat Transfer Characteristics of Nanofluids Flowing Across Pipe", 6th International Conference on Mechanical Engineering and Renewable Energy (ICMERE-2021), Chattogram, Bangladesh
- SI Alam, MS Islam, MR Halder, "Heat Transfer in Square Enclosure Filled with SWCNT Nanofluid With a Square Heat Source at the center", 4th International Conference On Innovations In Mechanical Engineering (ICIME-2021), Hyderabad, India

## Skills

**Research and Analysis:** LaVision DaVis (PIV), ImageJ (Pixel Analysis), DMC Motion Controller.

**CAD and Animation Tools:** Creo, SolidWorks, Siemens NX, AutoCAD, 3DS Max.

**Simulation Tools:** ANSYS, Abaqus, COMSOL.

**Programming Language:** MATLAB, Python.

**Documentation and Presentation Tool:** MS Office, LaTeX.

## Work Experience

**Graduate Teaching Assistant, Clarkson University – Potsdam, NY**

Aug 2023 – Present

- Mentored undergraduate students in Fluid Mechanics, Measurement & Instrumentation, Experimental Methods, and Test Engineering (ES 330, ME/AE 201/301/401) through in-class lectures and lab demonstrations, and provided individualized guidance during office hours and written feedback on coursework.

**Production Engineer, Bangladesh Steel Re-rolling Mills Ltd. – Chattogram, Bangladesh**

Mar 2023 – Jun 2023

- Designed and modified mechanical components (guides, vises) applying GD&T principles for precision fit and performance; contributed to a 4% reduction in operational delays through process and equipment optimization.

Certifications

- Lean Six Sigma Green Belt
  - Fundamentals of Engineering (FE) – Mechanical, Passed
  - Certified SOLIDWORKS Professional (CSWP)
  - Certified SOLIDWORKS Associate (CSWA)
- May 2025

Sep 2024

Apr 2021

Feb 2021

Academic Projects

- Unsteady CFD Simulation of Flow Around a Square Cylinder at Re = 100.**
- Jan 2026 - Jun 2026
- Developed a finite-volume CFD solver in MATLAB to simulate unsteady flow and vortex shedding around a square cylinder using pressure–velocity coupling and Rhie–Chow interpolation.
  - Evaluated drag and lift coefficients and compared predictions with commercial CFD results, analyzing the effects of mesh resolution and numerical schemes on solution accuracy.
- Deep Learning-Based Prediction of Li-Ion Battery Impedance Spectra.**
- Jan 2026 - Jun 2026
- Developed deep neural network regression models using TensorFlow/Keras to predict real and imaginary components of Li-ion battery impedance from state-of-charge, frequency, and battery test conditions.
  - Achieved R<sup>2</sup> values of approximately 0.97 and 0.96 for real and imaginary impedance, respectively, demonstrating a data-driven surrogate approach for generating EIS responses and expanding battery diagnostic datasets.
- Design and Manufacturing of a Spinning W-Electrode for GTAW Process (S-TIG).**
- Jan 2024 - Jun 2024
- Created detailed CAD models and assembly using SolidWorks of all the parts associated with the tungsten electrode.
  - Prepared project budget, bill of materials (BoM) and sourced materials for cost-effective production.
  - Coordinated with the machine shop for manufacturing and assembly, ensuring seamless communication among project members through regular updates and collaboration.
- Design & Animation of a Jet Engine.**
- Jan 2024 - Jun 2024
- Made CAD models of different parts of a jet engine and assembled using Creo.
  - Rendered and created an animation with 3DS Max.
- Numerical Simulation of Turbulent Flow in a Channel.**
- Jan 2024 - Jun 2024
- Simulated turbulent flow in rough channels using ANSYS Fluent, and validated predictions against experimental friction coefficients and boundary layer data.
- Finite Element Analysis of Airflow in a Constant Pressure Gradient Driven Duct.**
- Aug 2023 - Dec 2023
- Developed a 2D FEM model applying Galerkin method to analyze velocity profiles in an air duct under no-slip conditions.
  - Performed error and convergence analysis using Richardson Extrapolation, comparing structured and unstructured mesh results for accuracy.
- Computational Analysis of Fluid Dynamics: Electronic Cooling, Boundary Layer Behavior & Similarity Solution to Falkner-Skan Equation.**
- Aug 2023 - Dec 2023
- Analyzed boundary layer flow over a flat plate, electronic cooling in ducts using ANSYS Fluent.
  - Solved the Falkner-Skan equation using MATLAB to study velocity profiles and boundary layer thickness in wedge flow.

Standardized Test Results

GRE General Test Scores

|                        |                  |                    |
|------------------------|------------------|--------------------|
| Quantitative Reasoning | Verbal Reasoning | Analytical Writing |
| 164                    | 148              | 3.5                |

IELTS Academic Band Scores

|          |           |         |         |         |
|----------|-----------|---------|---------|---------|
| Speaking | Listening | Reading | Writing | Overall |
| 7.0      | 7.5       | 7.0     | 6.0     | 7.0     |